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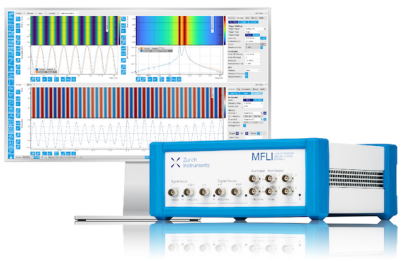
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An accurate determination of rovibrational spectra using the externally corrected coupled-cluster approaches: LiH ground state

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Recently acquired highly precise spectroscopic data for the ground state of LiH and its various isotopomers are employed to carry out a critical assessment of the performance of the externally corrected coupled-cluster (CC) approaches. Both the amplitude and energy-corrected approaches are considered, in particular the reduced multireference CC method with singles and doubles (RMR CCSD) and the asymmetric energy formula based CCSD-[MR] method, both exploiting the same modest-size multireference configuration interaction (MR CISD) wave function, based on an M-dimensional reference space, as the source of higher than pair clusters. To assess the size of the basis set errors relative to those of the methods employed, the comparison is also made with the full CI (FCI) results at the cc-pVTZ level. The rovibrational energy levels and the corresponding transition frequencies are then computed for various isotopomers of LiH using the theoretically determined potentials at the cc-pVXZ (X=D, T, Q, and 5) and the extrapolated complete-basis-set-limit levels and a comparison is made with the experiment as well as with the existing theoretical results, particularly those exploiting the CC approaches. The role of adiabatic corrections is also assessed. It is shown that both the RMR CCSD and CCSD-[4R] methods, yielding an almost indistinguishable results, are capable of accounting for the nondynamical correlation effects that are lacking in the standard single-reference CCSD approaches. © *2003 American Institute of Physics.* [DOI: 10.1063/1.1535438]

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I. INTRODUCTION

Lithium hydride, one of the lighter heteronuclear diatomics, continues to be the object of intense theoretical and experimental studies. The spectroscopic and structural data up to 1993 have been extensively reviewed by Stwalley and Zemke.[1] More recently, high-resolution spectra of LiH, recorded by the Bernath's group, provided even more accurate results.[2] These very precise experimental—or experimentally derived—data cover a wide range of internuclear separations and represent an ultimate benchmark that enables a very critical assessment of the performance and reliability of various theoretical approaches in describing a simple four-electron molecular system.

From the theoretical viewpoint, the simplicity of the electronic structure of LiH makes it an ideal molecule for testing of various techniques. Its electronic structure exhibits general features of more complex systems, including the role of triply and quadruply excited configurations in high-level correlated calculations. The fact that only four electrons are involved makes it possible to employ the full CI (FCI) approach with a fairly realistic basis set. This in turn enables a critical assessment of the performance of various methods relative to FCI, as well as an estimate of the basis set error by comparing the FCI results with highly accurate experimental data.

The number of papers reporting both the *ab initio* and semiempirical quantum mechanical calculations of various properties of LiH is very substantial. Since it is now well established that the reliable potential energy curves (PECs) for the entire range of geometries involving the breaking of true chemical bonds, which enable a computation of quantitatively correct spectroscopic data, can only be generated with multireference (MR) based methods that properly account for both the dynamic and nondynamic correlation effects and employ a sizable basis set, we mention in the following brief overview only highly correlated calculations of PECs, most of which also address the vibrational analysis. Already about 20 years ago, Partridge and Langhoff[3] reported the MCSCF-CI study using the Slater-type basis set. Their PEC is often used to this very day. About the same time, Jönsson *et al.*[4] carried out the MCSCF-CI calculations using a sizable Gaussian-type orbital basis set. More recently, Boutalib and Gadéa[5] computed both diabatic and adiabatic PECs, while relying on a nonempirical pseudopotential for the core electrons and the full CI treatment of valence electrons.

At the coupled-cluster level, Dykstra *et al.*[6,7] reported results obtained with the single-reference (SR) CC method involving singles and doubles and employing Brueckner orbitals (CCSD/BO), while Balková *et al.*[8] carried out state-universal MR CCSD[9] computations. However, in the latter work, the vibrational analysis was not carried out.

The vibrational analysis represents a very powerful tool for testing of the accuracy of the computed PECs. Generally, in all cases of highly accurate results,[3–5,7] the errors in com-

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puted vibrational term values steadily increase with the increasing vibrational quantum number. In other words, the theoretical PECs substantially deviate from the experimentally derived ones—often by several hundreds of wave numbers (cm^{-1})—for large internuclear separations R approaching the dissociation region. For this reason, some of the above mentioned studies focused only on lower-lying vibrational levels. Clearly, it is the outer region of the potential, describing the actual bond breaking, where the MR character of the wave function plays the most significant role. It is thus of interest to find out how well the CC-type methods (for an overview see, e.g., Refs. 10 and 11) are capable of describing this behavior.

In order to overcome the inability of the standard SR CCSD method to describe the multireference character of the wave function, we designed various *externally corrected* (ec) versions of this approach,[10,12–14] in particular the so-called *amplitude-correcting* reduced multireference (RMR) CCSD method,[15–17] as well as the *energy-correcting* CCSD-[MR] method.[18–22] In the former case, one employs a modest-size MR CISD wave function to estimate the three- and four-body connected clusters, which are then used to correct the standard SR CCSD equations. The latter approach then relies on the asymmetric energy formula and computes the CCSD energy by projecting onto a MR CISD wave function that employs an M-dimensional reference space. This produces an energy correction to the standard CCSD energy which accounts for the MR effects that are missing in the SR CCSD. The CCSD-[MR] method is closely related with the so-called *method of moments* CC theory[23–29] (cf. also Refs. 30 and 31). Remarkably enough, both the amplitude- and energy-corrected methods yield very similar results as long as the same MR CISD wave function is employed as the external source of nondynamic correlation effects.[18–22] For more details concerning the methodological aspects we refer the reader to the original papers.[15–22]

The purpose of this paper is to report on high-level CC calculations of the entire PEC for LiH and the subsequent computation of rovibrational levels and corresponding spectra. Since the standard SR CCSD method is likely to yield fairly accurate results for Li–H separations that do not excessively deviate from the equilibrium bond length, the MR effects will be primarily felt in the outer region of the PEC. This region represents the primary focus of our study. This is particularly important, since the MR CCSD study[8] does not seem to properly correct the shortcomings of the SR CCSD method in this range of geometries. Hence, it is of interest to examine the performance of the RMR CCSD and of the energy-corrected CCSD methods in this case.

II. METHOD AND COMPUTATIONAL DETAILS

With the exception of the FCI results, all the methods that are employed in this study are based on the exponential CC Ansatz for the wave function. In addition to the standard SR CCSD method, we employ both the ''amplitude-corrected'' RMR CCSD[15] and the ''energy-corrected'' CCSD[18,19,22] methods. The details of these methods, as well as their characteristics and ability to handle nondynamic correlation effects that are lacking in the SR CCSD approach, have been amply described in our earlier papers,[15,18,19,22] and need not be repeated here.

Since LiH is single bonded, the multireference effects arising in the dissociation regime are due to the orbital degeneracy of the σ bonding and σ^* antibonding orbitals. We hence use the four-reference (4R) model space for the required MR CISD. This model space is spanned by the single determinantal Hartree–Fock reference, and by one doubly and two singly excited configurations involving the σ bonding and σ^* antibonding orbitals. Following the convention adopted in Ref. 18, the corresponding methods are designated by the acronyms 4R-RMR CCSD (or, simply, 4R-RMR) and CCSD-[4R], respectively.

The *ab initio* models that we employ are defined by the correlation-consistent (cc) basis sets cc-pVXZ (X=D, T, Q, and 5).[32,33] When not indicated otherwise, Cartesian Gaussians are used. To generate the restricted Hartree–Fock MOs, as well as the necessary one- and two-electron integrals, we employ the GAMESS package.[34] All calculations concern the electronic ground state of the LiH molecule, and all electrons are correlated in all CC or CI calculations.

The rovibrational energy levels $E(\nu,J)$ associated with the vibrational quantum number ν and the rotational quantum number J have been determined via the numerical integration of the radial Schrödinger equation for the nuclear motion

$$\left\{\frac{\hbar^2}{2\mu}\frac{d^2}{dR^2}-U^{\mathrm{eff}}(R)-\frac{\hbar^2}{2\mu R^2}[J(J+1)-\Omega^2]+E(\nu,J)\right\}\psi(R;\nu,J)=0, \tag{1}$$

where μ designates the reduced mass of the system, Ω is the projection of the electronic angular momentum onto the internuclear axis, and $U^{\mathrm{eff}}(R)$ represents the rotationless effective internuclear potential. In our case of a $^1\Sigma_g$ state, $\Omega=0$. The radial Schrödinger equation is solved using the LEVEL codes of LeRoy.[35]

The effective potential $U^{\mathrm{eff}}(R)$ is given by the sum of mass-independent, clamped-nuclei (CN), or Born–Oppenheimer, potential U^{CN} and an adiabatic correction term ΔU^{Ad} that accounts for the Born–Oppenheimer breakdown

$$U^{\mathrm{eff}}(R)=U^{\mathrm{CN}}(R)+\Delta U^{\mathrm{Ad}}(R). \tag{2}$$

For the diatomic LiH molecule, the adiabatic correction term ΔU^{Ad}, Eq. (2), can be further broken up into the components associated with the Li and H atoms, namely

$$\Delta U^{\mathrm{Ad}}(R)=\Delta U^{\mathrm{Li}}_{\mathrm{Ad}}(R)/M_{\mathrm{Li}}+\Delta U^{\mathrm{H}}_{\mathrm{Ad}}(R)/M_{\mathrm{H}}, \tag{3}$$

where $\Delta U^{\mathrm{A}}_{\mathrm{Ad}}(R)$ (A=Li and H) are mass-invariant adiabatic correction functions.

Several studies[2,36–39] have been concerned with the role of adiabatic corrections. In general, the magnitude of these corrections is small, although—in percentage terms—one finds significantly different estimates. Nonetheless, relative to the clamped-nuclei potential, the adiabatic corrections are very small. To assess the effect of these corrections on

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TABLE I. The FCI total energy (in a.u.) and the CCSD, 4R-RMR, and CCSD-[4R] energies relative to the FCI one (in mhartree) of ^{7}LiH for a selected subset of internuclear separations R (in Å), as obtained with the cc-pVTZ basis set. The last two rows give the nonparallelism error (NPE) of the potential energy curve relative to the FCI curve. See the text for details.

R	FCI	CCSD	4R-RMR	CCSD-[4R]
1.0	−7.945 930	0.183	0.173	0.173
1.3	−8.025 048	0.124	0.116	0.116
1.6	−8.040 025	0.105	0.096	0.096
1.9	−8.032 129	0.102	0.090	0.090
2.2	−8.017 189	0.110	0.094	0.094
2.5	−8.001 057	0.131	0.106	0.105
2.9	−7.981 700	0.185	0.137	0.134
3.5	−7.961 755	0.359	0.216	0.204
4.1	−7.952 988	0.604	0.264	0.248
4.7	−7.950 204	0.784	0.241	0.231
6.6	−7.949 109	0.976	0.122	0.117
NPE (mhartree)		0.875	0.174	0.158
NPE (cm^{-1})		192.0	38.2	34.7

computed rovibrational energy levels, we have chosen the adiabatic corrections derived by Chan *et al.*,[36] since they are closely related with the experiment-based vibrational energies derived via the inverted perturbation approach[40–42] (IPA) that we employ in this paper for a comparison. For the H atom, the $\Delta U^{\mathrm{H}}_{\mathrm{Ad}}$ term is tabulated in Ref. 36, while the contribution of the Li atom to the adiabatic correction function ΔU^{Ad} for the ground-state potential energy is deemed to be insignificant.[36] Since the $\Delta U^{\mathrm{H}}_{\mathrm{Ad}}$ function is tabulated for a different set of internuclear separations than those used in our computations of PECs, we have employed a polynomial fit for the $\Delta U^{\mathrm{H}}_{\mathrm{Ad}}(R)$ term, which was then added to our *ab initio* clamped-nuclei potential. Thus, for every internuclear separation R, for which we determined our potential, we have chosen a few points in the vicinity of this R value from the $\Delta U^{\mathrm{H}}_{\mathrm{Ad}}$ table,[36] fitted them to a suitable polynomial, and used the interpolated value of the adiabatic correction term. In the following, both results, obtained with and without the adiabatic corrections, are presented.

III. RESULTS

A. Comparison with the FCI benchmark

We first consider the FCI calculations using the cc-pVTZ basis set. For the four electron LiH molecule, this basis set is quite large and theoretical results should provide a reasonable description that allows a meaningful comparison with experiment.

A representative sample of the FCI total energies, and of the deviations of the CCSD, 4R-RMR, and CCSD-[4R] energies from the FCI ones, for 12 internuclear separations R selected from 41 geometries considered, is given in Table I. A rough measure of the adequacy of approximate PECs is afforded by the so-called nonparallelism error (NPE),[10] defined as the difference between the signed maximal and minimal deviations from the FCI potential within a chosen interval of internuclear separations. The NPEs for the CCSD, 4R-RMR, and CCSD-[4R] potentials are listed in the last row of Table I. These NPEs have been calculated using all 41 geometries considered and should accurately reflect the degree of "parallelism" between the approximate and FCI PECs within the considered region of internuclear separations R ranging from 1 to 6.6 Å . As will be seen later on, these NPEs closely reflect the deviations in computed vibrational term values.

The deviations of the 4R-RMR and CCSD-[4R] energies from the FCI ones are rather small and never exceed 0.3 millihartree (mhartree). In contrast, the CCSD errors approach a 1 mhartree level for highly stretched geometries. Nonetheless, even the CCSD energies do not deviate by more than 0.3 mhartree from the FCI ones for the internuclear separations not exceeding two-times the equilibrium bond distance R_e. This range corresponds to about 75% of the potential energy well. Thus, the CCSD method is sufficiently accurate for the description of at least the bottom half of the vibrational levels. The multireference effects become essential only when approaching the dissociation limit and thus affect only a few highest-lying vibrational levels.

At this point it is interesting to compare the above presented results with the state-universal MR CCSD[9] ones obtained by Balková *et al.*[8] Even though we cannot draw quantitative conclusions from this comparison, since in each case a different basis set was employed, we can get some idea of the basic characteristics of the methods involved. For the corresponding geometries, the MR CCSD method using the Hartree–Fock orbitals yields either very similar or slightly worse results than does the SR CCSD method. Specifically, the MR CCSD errors increase from less than 0.2 mhartree at R_e to more than 1 mhartree at the outer end of the geometries considered. As implied by our results, this is precisely the same behavior that characterizes the standard CCSD approach, which should have been corrected when the MR-type CC method is used.

Although this undesirable performance of MR CCSD can be avoided by relying on the quasirestricted HF (QRHF) orbitals rather than on the ubiquitous Hartree–Fock ones, another drawback arises in that the QRHF based MR CCSD equations cease to converge when the LiH internuclear separation R exceeds 9 bohr, i.e., as the dissociation limit is approached. This failure is due to the ionic-covalent avoided crossing between the ground $X\,^1\Sigma^+$ and the first excited $A\,^1\Sigma^+$ states. As may be seen from Table I of Ref. 8, the errors of the QRHF based MR CCSD method steadily increase for the $A\,^1\Sigma^+$ state, until the algorithm for solving of the equations no longer converges. This example thus indicates the methodological and computational difficulties that we face when employing the genuine MR CCSD methods, in this case of the state-universal,[9] multireference, multistate type. On the other hand, the RMR CCSD method that is employed here represents a state-specific or state-selective approach, which may also be characterized as a one-state multireference-type method that avoids the intruder state problems of multistate approaches.

Next, we present in Table II all 23 vibrational energy levels as obtained with the FCI/cc-pVTZ potential, as well as the differences from this FCI benchmark when employing the CCSD, 4R-CISD, 4R-RMR, and CCSD-[4R] potentials. We observe that (i) the CCSD method yields very precise term values for the bottom half of the vibrational manifold,

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TABLE II. The FCI vibrational levels for ^{7}LiH and the deviations of the CCSD, 4R-CISD, 4R-RMR CCSD, and CCSD-[4R] vibrational levels from the FCI ones, obtained with the cc-pVTZ basis set. The last row lists the absolute mean error $|\Delta|_{av}$. All values are in cm^{-1}.

ν	FCI	CCSD	4R CISD	4R-RMR	CCSD-[4R]
0	701.9	0.1	1.2	0.0	0.0
1	2069.3	0.2	4.3	0.1	0.0
2	3388.5	0.5	8.1	0.2	0.2
3	4662.3	0.8	12.4	0.4	0.3
4	5892.8	1.3	17.4	0.6	0.6
5	7081.6	1.8	23.0	0.9	0.9
6	8229.5	2.5	29.5	1.3	1.2
7	9337.1	3.3	36.8	1.8	1.6
8	10 404.6	4.2	45.2	2.3	2.1
9	11 432.1	5.3	54.8	3.0	2.8
10	12 418.9	6.7	65.7	3.7	3.5
11	13 364.4	8.4	78.4	4.6	4.3
12	14 267.3	10.5	93.1	5.8	5.3
13	15 125.8	13.1	110.3	7.1	6.6
14	15 937.2	16.4	130.4	8.8	8.1
15	16 698.0	20.6	154.2	10.8	9.8
16	17 403.4	26.2	182.3	13.2	12.0
17	18 046.8	33.7	215.6	16.0	14.6
18	18 619.8	43.8	254.7	19.4	17.6
19	19 111.4	57.8	299.5	23.0	20.9
20	19 507.3	77.5	347.6	26.3	23.9
21	19 790.2	105.9	386.5	27.2	25.0
22	19 939.7	148.4		24.5	23.4
$\|\Delta\|_{av}$		25.6	116.0	8.7	8.0

TABLE III. Experimentally based IPA vibrational energy levels (Ref. 1) of ^{7}LiH and the differences between the FCI, CCSD, and 4R-RMR computed values and the experimental ones, as obtained using the cc-pVTZ basis. The last row lists the absolute mean error $|\Delta|_{av}$. All values are in cm^{-1}.

ν	Expt. [a]	FCI	CCSD	4R-RMR
0	697.9	4.0	4.1	4.0
1	2057.6	11.7	11.9	11.8
2	3372.5	16.0	16.5	16.2
3	4643.4	18.9	19.8	19.3
4	5871.1	21.7	22.9	22.3
5	7056.6	25.0	26.8	25.9
6	8200.4	29.1	31.6	30.4
7	9303.0	34.1	37.4	35.9
8	10 364.7	39.9	44.1	42.2
9	11 385.9	46.2	51.5	49.1
10	12 366.4	52.5	59.2	56.2
11	13 306.0	58.4	66.8	63.0
12	14 204.1	63.2	73.7	69.0
13	15 059.6	66.2	79.2	73.3
14	15 870.8	66.4	82.8	75.2
15	16 635.2	62.8	83.4	73.5
16	17 349.5	53.9	80.1	67.0
17	18 008.7	38.1	71.8	54.1
18	18 606.6	13.2	57.0	32.6
19	19 134.5	−23.1	34.7	−0.1
20	19 581.1	−73.8	3.7	−47.5
21	19 932.1	−141.9	−36.0	−114.7
22	20 169.8	−230.1	−81.7	−205.6
$\|\Delta\|_{av}$		51.7	46.8	51.7

[a]Reference 1.

in which case the errors do not exceed 10 cm^{-1}. However, for higher-lying levels, the errors become considerably larger. In particular, for the top two levels, the CCSD errors exceed 100 cm^{-1} and the mean deviation is 25.6 cm^{-1}. (ii) The 4R-RMR and CCSD-[4R] results are very similar to one another and most satisfactory. In both cases the mean deviation is less than 10 cm^{-1} for all the levels. Although the 4R-RMR and CCSD-[4R] term values for the lower-lying levels differ little from the CCSD ones, their superiority is indisputable for the higher-lying vibrational levels. Finally, (iii) the 4R-CISD results themselves are not satisfactory, although the corresponding wave function serves us well as the external source for an estimate of the higher-than-pair clusters.

It is worthwhile to observe that the maximum deviation between the approximate and FCI vibrational term values (cf. Table II) nicely correlates with the NPE of the corresponding PEC (cf. Table I), as already alluded to above. Indeed, in the case of the CCSD, 4R-RMR, and CCSD-[4R] methods, these maximal deviations (148.4, 27.2, and 25.0 cm^{-1}, respectively) amount to roughly three quarters (i.e., 71%–77%) of the corresponding NPE values (192, 38.2, and 34.7 cm^{-1}, respectively). Thus, the NPE provides us with a rather good, yet simple, single-parameter, measure of the quality of the computed PECs.

In Table III we also compare the vibrational energy levels obtained with the FCI, CCSD, and 4R-RMR potentials at the cc-pVTZ level with the experimental values derived via the inverted perturbation approach (IPA).[40–42] These experimentally derived term values are based on the total effective internuclear potential that also accounts for the Born–Oppenheimer breakdown. However, the latter represents only a very small correction. Both 4R-CISD and CCSD-[4R] results are omitted, since the former one yields very poor results (cf. also Table II), while the latter one gives an almost identical results as does the 4R-RMR CCSD method.

Now, the FCI values represent the exact result for the cc-pVTZ *ab initio* model. Comparing the discrepancies between the approximate and FCI term values given in Table II with those of Table III, we must conclude that the basis set error clearly exceeds the errors of the externally corrected CCSD methods. Indeed, in the 4R-RMR case, the average error relative to FCI, representing the error of the method, amounts to 8.7 cm^{-1}, while the average basis set error, implied by the differences between the FCI and experimental vibrational term values, is 51.7 cm^{-1}.

It is also important to notice that the deviations of the FCI and 4R-RMR term values from the experimental ones are very similar, as implied by the proximity of the FCI and 4R-RMR results (cf. Table II), while the corresponding deviations for the standard SR CCSD method are slightly ''inferior'' (cf. the average discrepancies given in Table III, i.e., 51.7 cm^{-1} for both FCI and 4R-RMR and 46.8 cm^{-1} for SR CCSD). This behavior is typical when one employs not sufficiently large basis sets, as thoroughly demonstrated in Ref. 22 for the HF molecule. Indeed, the SR CCSD results are usually ''superior'' to the higher-level RMR CCSD or CCSD-[MR] ones when using a smaller basis set (e.g., cc-pVDZ), while the opposite is the case once we approach the complete basis set (cbs) limit. We shall observe the same behavior in the present case as well, even though in this

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TABLE IV. Experimentally determined fundamental ($\nu \rightarrow \nu+1$) vibrational transition energies[1] [Expt.(IPA)] for ^{7}LiH and the deviations of the computed values from the experimental ones as obtained with various methods. All values are in cm^{-1}.

ν	Expt.(IPA)[a]	CCSD cc-pVQZ	CCSD-[4R] cc-pVQZ	CI/STF[b]	CI[c]	CCSD/ BO[d]	MCSCF/ CI[e]
0	1359.7	2.0	1.8	−6.8	−14.9	−1.7	−9.6
1	1314.9	0.8	0.5	−5.9	−14.6	−2.2	−8.8
2	1270.9	0.5	0.2	−4.4	−14.2	−1.8	−7.9
3	1227.8	0.8	0.5	−4.2	−12.8	−0.5	−7.2
4	1185.4	1.5	1.2	−5.3	−10.8	2.0	−6.9
5	1143.8	2.2	1.9	−8.2	−8.8	5.1	−7.1
6	1102.6	2.9	2.5	−11.2	−7.2	8.8	−7.5
7	1061.8	3.5	3.0	−10.8	−6.3	12.2	−8.2
8	1021.2	4.0	3.4	−11.6	−6.0		−8.5
9	980.5	4.5	3.7	−10.6	−5.7		−9.8
10	939.6	4.8	3.8	−10.2	−5.4		
11	898.1	5.0	3.8	−9.8	−5.1		
12	855.5	5.2	3.7	−10.0	−4.5		
13	811.2	5.4	3.3	−11.2	−3.7		
14	764.4	5.5	2.7	−12.6	−2.8		
15	714.2	5.5	1.7	−14.2	−1.9		
16	659.3	5.5	0.0	−16.2	−1.1		
17	597.9	5.5	−2.4	−19.4	−0.4		
18	527.9	5.8	−6.0	−24.8	0.1		
19	446.6	6.9	−11.4	−31.4	0.3		
20	351.0	9.3	−20.2	−39.0	−0.6		
21	237.7	15.4	−36.1	−48.2	−2.0		

[a]Reference 1.
[b]Reference 3.
[c]Reference 5.
[d]Reference 7.
[e]Reference 4.

four-electron case the SR CCSD method works extremely well, with a possible exception of a few highest-lying vibrational levels.

B. Vibrational levels and transition energies

In Table IV we present the experimentally directly observable values of the fundamental ($\Delta\nu=1$) vibrational transition energies $\Delta G_\nu \equiv G_{\nu+1} - G_\nu$ and compare our results with both the experiment and the results obtained by other authors. The experimental data represent the differences between the IPA energies,[1] and the theoretical results are presented as deviations from the experimental values. The transition energies or the corresponding frequencies reflect the energy spacing between the neighboring vibrational levels, and thus characterize the directly observed experimental spectrum. Clearly, they are independent of the ground-state energy $E(0,0)$ and have been most often used for comparison in previous publications by other researchers.

We see clearly from Table IV that the CCSD/cc-pVQZ and CCSD-[4R]/cc-pVQZ vibrational transition energies ΔG_ν are among the best. The 4R-RMR/cc-pVQZ results have not been included, since they are almost the same as the CCSD-[4R] ones. Our CCSD results are also very close to those obtained by Lee *et al.*[7] with the CCSD/BO method which effectively eliminates one-body clusters.

It is interesting that in the case of the CI-type approaches[3–5] that are included in Table IV, almost all deviations of the computed transition energies are negative, i.e., the computed spacings are too small. Since the limited CI-type approaches are primarily lacking the disconnected triples and quadruples that are well accounted for by the CC-type methods, we can speculate that these higher-than-pair clusters are responsible for the observed increase in the energy spacings.

Even when the errors in the individual energy spacings are small, amounting to a few wave numbers, they can accumulate to hundreds of wave numbers when considering the term values for higher vibrational levels. Since most previous studies[3–5,7] do not give the actual vibrational energies, we present in Table V the cumulative errors obtained as the sum of the errors in the transition energies up to a given level, which provide us with a measure of the error in the corresponding vibrational energy term. For example, the error of Partridge and Langhoff's result[3] in the $\nu=9$ vibrational energy amounts to about -80 cm^{-1}, while the CCSD/cc-pVQZ and CCSD-[4R]/cc-pVQZ errors are about 20 cm^{-1}, etc. Again, in view of this comparison, our results belong to the most accurate ones.

Clearly, the cumulative errors (or the errors in the vibrational term values) generally increase with the increasing vibrational quantum number, since the corresponding outer turning points correspond to highly stretched geometries for which the computation of the potential is most challenging. This is also the case for the CCSD method: A few highest-lying CCSD/cc-pVQZ levels deviate from the experiment by about 100 cm^{-1}, and the same trend is implied by the avail-

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TABLE V. Cumulative errors in computed vibrational energy levels of ^{7}LiH as determined from the errors in fundamental transition energies of Table IV for the indicated range of vibrational quantum numbers ν. All values are in cm^{-1}.

ν	CCSD cc-pVQZ	CCSD-[4R] cc-pVQZ	CI/STF[a]	CI[b]	CCSD/ BO[c]	MCSCF/ CI[d]
0–4	5.6	4.2	−26.6	−67.3	−4.2	−40.4
0–7	14.2	11.6	−56.8	−89.6	21.9	−63.2
0–9	22.7	18.7	−79.0	−101.3		−81.5
0–14	48.6	36.0	−132.8	−122.8		
0–19	77.8	17.9	−238.8	−125.8		
0–21	102.5	−38.4	−326.0	−128.4		

[a]Reference 3.
[b]Reference 5.
[c]Reference 7.
[d]Reference 4.

able CCSD/BO results, in which case the cumulative errors are even larger.

Thus, the comparison of the computed and experimental vibrational term values represents a more severe test of theoretical methods than does the comparison of vibrational frequencies. Such a comparison for ^{7}LiH, as obtained with the 4R-CISD, CCSD, 4R-RMR, and CCSD-[4R] methods, is presented in Table VI. To facilitate this comparison, all theoretical results are presented as deviations from the experimental (IPA) data that are listed in Table III. In terms of mean deviations, the 4R-RMR or CCSD-[4R] errors amount to about one-half of the CCSD ones. Although the CCSD method provides very good energies for the bottom half of the vibrational levels, the deviations from the experiment reach almost 100 cm^{-1} for the highest-lying levels. This is clearly due to the multireference character of the ground state in the dissociation limit and this shortcoming can be to a large extent rectified by relying on the (amplitude-corrected) 4R-RMR or (energy-corrected) CCSD-[4R] approaches. Indeed, the 4R-RMR and CCSD-[4R] errors are considerably smaller than the CCSD ones, and are likely due to—in an equal measure—the intrinsic error of the method and the basis set error, since the former one does not exceed 10 cm^{-1} as the results given in Table II indicate. Furthermore, the CCSD[4R]/cc-pVQZ potential is too low and, consequently, the computed vibrational energy levels show a negative deviation. This deficiency can only be overcome by exploiting a very large basis set. However, when we employ the largest available cc-pV5Z basis,[33] we encounter convergence problems (at both the CI and CC levels) for small internuclear separations R ($R<\sim 1$ Å) and for severely stretched geometries ($R>\sim 4$ Å). For the indicated regions of internuclear separations, even the standard CISD from GAMESS does not converge. Hence, we can only obtain a partial PEC, covering roughly 80% of the potential well, and perform only the 4R-RMR CCSD calculations in this case. The resulting partial PEC can be nonetheless used to generate the first 19 vibrational levels, which are given in Table VI. A few highest-lying levels (for which the values in Table

TABLE VI. The deviation of the computed vibrational energy levels as obtained with various methods from the experimental (IPA) ones for ^{7}LiH. The CCSD, 4R-CISD, 4R-RMR, and CCSD-[4R] values in columns 2–5 were obtained with the cc-pVQZ basis set, while those in the second last column were obtained with the 4R-RMR potential extrapolated to the complete basis set limit (cbs). The deviations in the last column correspond to the CCSD-[4R]/cc-pVQZ potential including the adiabatic corrections. The last row lists the absolute mean deviations $|\Delta|_{av}$. All values are in cm^{-1}. See the text for details.

ν	CCSD	4R-CISD	4R-RMR	CCSD-[4R]	4R-RMR cc-pV5Z	4R-RMR cbs	CCSD-[4R] cc-pVQZ + Ad.
	cc-pVQZ						
0	0.4	3.3	0.4	0.4	−0.2	−0.6	0.2
1	2.4	11.9	2.2	2.2	1.0	−1.8	1.6
2	3.2	20.2	2.8	2.8	1.3	−3.0	1.7
3	3.6	29.1	3.0	3.0	1.8	−3.9	1.6
4	4.4	39.3	3.6	3.5	2.8	−4.1	1.7
5	5.9	51.3	4.8	4.7	4.2	−4.0	2.5
6	8.2	65.3	6.7	6.6	5.9	−3.5	4.0
7	11.1	81.4	9.2	9.1	7.7	−2.9	6.2
8	14.6	99.6	12.3	12.1	9.6	−2.4	8.8
9	18.6	120.2	15.8	15.5	11.8	−1.6	11.9
10	23.1	143.3	19.5	19.1	14.6	−0.3	15.2
11	27.9	169.1	23.4	22.9	17.5	1.3	18.6
12	32.9	198.1	27.4	26.8	20.7	3.3	22.1
13	38.2	230.6	31.3	30.5	24.0	5.8	25.4
14	43.6	267.1	34.9	33.8	27.6	9.1	28.4
15	49.1	308.4	37.9	36.5	31.2	13.0	30.6
16	54.6	355.1	40.0	38.2	(35.1)	18.1	31.9
17	60.0	408.0	40.7	38.2	(39.2)	24.3	31.4
18	65.5	467.9	39.0	35.9	(40.0)	28.4	28.4
19	71.4	534.9	33.7	29.9			21.8
20	78.2	607.5	22.9	18.4			9.6
21	87.6	704.9	2.8	−1.7			−11.4
22	102.9	467.2	−34.4	−37.8			−48.5
$\|\Delta\|_{av}$	35.1	234.1	19.5	18.7			15.8

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TABLE VII. Differences ΔE between the SCF, CCSD, and 4R-RMR energies of ^{7}LiH (in mhartree) obtained with spherical (sph) and Cartesian (car) cc-pVQZ basis sets for selected internuclear separations R (in Å). The last two columns show the corresponding relative CCSD and 4R-RMR energy differences between the shifted spherical and Cartesian potentials so that they coincide at the equilibrium geometry.

	$\Delta E \equiv E_{\text{sph}} - E_{\text{car}}$			set $\Delta E(1.6) = 0.0$	
R (Å)	SCF	CCSD	4R-RMR	CCSD	4R-RMR
1.0	0.07	11.36	11.36	−3.18	−3.18
1.3	0.06	13.37	13.37	−1.17	−1.17
1.6($\sim R_e$)	0.05	14.54	14.54	0.00	0.00
2.4	0.04	15.01	15.01	0.47	0.47
3.2	0.10	15.47	15.48	0.93	0.94
4.8	0.36	15.37	15.50	0.83	0.96
6.4	0.63	15.31	15.52	0.77	0.98

TABLE VIII. The deviations of the computed vibrational energy levels of ^{7}LiH from the experimentally determined values (in cm^{-1}) as determined with the CCSD and 4R-RMR methods and spherical cc-pVQZ basis set (cf. with Table VI).

ν	CCSD	4R-RMR	ν	CCSD	4R-RMR
0	−6.6	−6.7	12	79.3	74.6
1	−14.8	−15.1	13	94.5	88.6
2	−23.5	−23.8	14	109.7	102.3
3	−25.8	−26.3	15	125.1	115.6
4	−22.4	−23.2	16	140.7	128.2
5	−15.0	−16.0	17	156.5	140.0
6	−4.9	−6.1	18	172.8	150.4
7	7.2	5.6	19	190.0	158.8
8	20.5	18.5	20	208.8	163.8
9	34.6	32.1	21	230.6	163.5
10	49.2	46.2	22	258.6	153.2
11	64.2	60.4			

VI are enclosed in parentheses) correspond to the region of internuclear separations for which we encounter the above mentioned convergency problems. This partial PEC is also sufficient for the generation of all the rovibrational levels that are required for the comparison with recently obtained highly accurate spectra[2] (cf. Sec. III F).

C. The role of Cartesian versus spherical basis functions

The results presented in the preceding sections (Secs. II A and II B) have been obtained with Cartesian Gaussians. In order to reduce the computational effort, one often represents the angular component via spherical harmonics, leading to the spherical Gaussians. This eliminates the ''contamination'' of Cartesian atomic orbitals (AOs) by the orbitals having a lower angular momentum quantum number and results in a considerably smaller orbital space, particularly for basis sets involving *f* and *g* orbitals. In this section we briefly point out that the use of Cartesian versus spherical basis sets can lead to significant differences when computing the entire PECs and the corresponding rovibrational levels.

In Table VII we present the differences between the energies obtained with Cartesian and spherical basis functions of the same cc-pVQZ basis set for a series of intermolecular separations R. At the SCF level, these differences are very small, while at the correlated level, represented here by the CCSD and 4R-RMR CCSD results, the ''Cartesian PEC'' lies about 11–15 mhartree below the corresponding ''spherical PEC.'' More importantly, however, these two PECs are not ''parallel'' to one another. Shifting both potentials so that they coincide at 1.6 Å ($\sim$R$_e$), the spherical PEC is above the Cartesian one for $R > R_e$ and below it for $R < R_e$. This results in a NPE error of $\sim$4 mhartree, i.e., $\sim$900 cm^{-1}. This NPE is large enough to shift the vibrational energy levels by $\sim$100 cm^{-1}. To illustrate this fact, we present in Table VIII the deviations of computed CCSD and 4R-RMR vibrational energies from experiment as obtained with the cc-pVQZ spherical Gaussians. A comparison with the corresponding results generated with the Cartesian Gaussians (Table VI) reveals that those obtained with the spherical Gaussians are systematically inferior, the discrepancies at the 4R-RMR level exceeding 100 cm^{-1} for $\nu \geqslant 14$.

With the largest available cc-pV5Z basis set we encounter unexpected predicaments. As already mentioned in the preceding section, with the Cartesian cc-pV5Z basis we encounter convergence problems for certain internuclear separations. In contrast, no convergence problems arise with the spherical cc-pV5Z basis. Yet, the corresponding PEC lies at higher energies than does the Cartesian PEC obtained with the cc-pVQZ basis set, and the computed vibrational energy levels are very close to those generated with the spherical cc-pVQZ basis and thus inferior to those realized with the Cartesian cc-pVQZ basis.

Moreover, the spherical Gaussian PECs show abnormal convergence behavior once we reach the quintuple-zeta level: The average energy lowering when proceeding from the X=Q to X=5 basis is considerably larger and out of step with the corresponding lowering when going from the X=T to X=Q basis. Thus, the energy differences between the X=D and T, T and Q, and Q and 5 results are 21.9, 5.9, and 10.4 mhartree at 1.6 Å ($\sim R_e$) and 14.4, 4.1, and 11.4 mhartree at $R = 5$ Å. Consequently, any extrapolation to the complete basis set limit (cf. Sec. III E) that employs spherical Gaussian PECs will lead to spurious results, as long as the cc-pV5Z PEC is included. However, no such problems arise when the Cartesian PECs are used (cf. Sec. III E).

In summary, the discrepancies between the computed potentials obtained at a highly correlated level when using Cartesian or spherical Gaussians involving *f* and *g* functions may be quite significant. Our experience indicates—at least in the case of systems involving the lightest first row (i.e., Li) atom—that the Cartesian basis sets provide more reliable and physically meaningful results.

D. Adiabatic corrections

By adding the adiabatic correction to the clamped nuclei or Born–Oppenheimer potential, we obtain the mass-dependent adiabatic potential as outlined in Sec. II. Using this potential we have calculated the corrected vibrational term values and the results are shown in the last column of Table VI. We present these results only for the CCSD-[4R]/cc-pVQZ potential, since the 4R-RMR results are very similar. In fact, except for the top two levels, the CCSD-[4R]

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TABLE IX. The deviations of the 4R-RMR CCSD transition frequencies, as obtained with the cc-pVXZ (X =D, T, Q, and 5) basis sets, from the experimental values (Ref. 2) (Expt.) for the selected lines in the P and R branches of various vibrational bands of ^{7}LiH. In each branch, the observed lines associated with the smallest and the largest rotational quantum number J'' were chosen. All values are in cm^{-1}. See the text for details.

Band	Line	Expt.[a]	X:=D	T	Q	5	5(Ad)[b]
(1,0)	P(28)	858.1	−9.0	1.2	3.5	0.2	0.1
	P(1)	1344.9	−25.5	7.7	1.9	1.2	0.7
	R(0)	1374.1	−25.9	7.8	1.8	1.1	0.7
	R(22)	1550.1	−22.5	4.4	2.5	0.1	−0.4
(2,1)	P(27)	844.9	−10.4	2.0	2.9	1.2	1.1
	P(1)	1300.5	−23.5	4.4	0.6	0.4	0.0
	R(0)	1328.8	−23.8	4.5	0.6	0.4	0.0
	R(17)	1485.9	−23.0	3.6	1.1	0.4	−0.2
(3,2)	P(23)	885.7	−13.3	2.9	2.4	1.8	1.6
	P(1)	1256.9	−21.9	3.1	0.3	0.5	0.1
	R(0)	1284.5	−22.3	3.0	0.2	0.5	0.1
	R(18)	1439.4	−22.1	3.3	1.2	1.2	0.7
(4,3)	P(21)	888.2	−15.0	3.7	2.4	2.1	1.9
	P(2)	1200.3	−20.7	3.0	0.6	1.0	0.6
	R(1)	1253.7	−21.4	3.0	0.5	1.0	0.6
	R(14)	1373.1	−22.0	3.7	1.3	1.5	1.0
(5,4)	P(15)	957.1	−17.5	4.2	2.1	1.9	1.7
	P(3)	1144.9	−20.1	3.6	1.2	1.4	1.1
	R(4)	1244.9	−21.3	3.7	1.2	1.4	1.0
	R(8)	1283.8	−21.7	4.0	1.5	1.6	1.1
(2,0)	P(25)	2109.9	−27.6	5.1	5.8	1.2	0.7
	P(1)	2659.8	−49.2	12.1	2.4	1.6	0.7
	R(1)	2701.6	−49.6	12.3	2.5	1.5	0.6
	R(13)	2785.8	−46.2	10.6	3.5	1.2	0.2
(3,1)	P(22)	2116.9	−31.3	5.8	4.2	2.5	1.9
	P(1)	2571.4	−45.5	7.5	0.9	0.9	0.1
	R(0)	2599.3	−45.9	7.5	0.9	0.8	0.0
	R(11)	2689.0	−44.6	7.1	1.5	1.1	0.3
(4,2)	P(19)	2118.4	−34.1	6.7	3.7	3.3	2.7
	P(3)	2454.5	−42.5	6.0	0.8	1.5	0.8
	R(1)	2524.2	−43.4	6.1	0.7	1.5	0.7
	R(11)	2598.1	−42.8	6.4	1.6	2.1	1.2
(5,3)	P(16)	2114.2	−36.5	7.8	3.8	3.6	3.1
	P(2)	2385.4	−41.3	6.5	1.7	2.4	1.6
	R(5)	2477.1	−42.2	6.9	2.0	2.5	1.7
	R(13)	2511.6	−42.0	7.9	3.2	3.3	2.5
(6,4)	P(11)	2145.4	−39.2	8.8	4.0	3.6	3.0
	P(5)	2255.9	−40.5	8.3	3.3	3.2	2.5

[a]Reference 2.
[b]The 4R-RMR/cc-pV5Z potential including the adiabatic corrections.

results are marginally better than the 4R-RMR ones. Since only the adiabatic correction function for the hydrogen atom is available[36] (the correction due to the heavier Li atom being deemed negligible), the results given in Table VI account only for the adiabatic correction due to the H atom.

The tabulated $\Delta U_{\mathrm{Ad}}^{\mathrm{H}}$ adiabatic correction function of Ref. 36 is largest (30–40 cm^{-1}) for short internuclear separations R, passes through zero at the equilibrium bond distance R_e, and levels off at -12 cm^{-1} at the outer end of the potential. It influences the computed vibrational energy levels by only a few wave numbers. The largest difference between the uncorrected and corrected term values occurs for the top level, and amounts to 10.7 cm^{-1} when using the CCSD[4R]/cc-pVQZ potential, i.e., roughly the value of the adiabatic correction function at large R. Since the adiabatic correction always decreases the vibrational energies, its account always improves the vibrational term values, with the exception of the top two levels for which we find a negative deviation from the experimental value already with an uncorrected potential. On the average, the adiabatic corrections amount to about 3 cm^{-1}.

E. Complete basis set limit

Similarly as in our earlier work on the HF molecule,[22] we have attempted to extrapolate the cc-pVXZ (X = D, T, Q, and 5) results towards the complete basis set (cbs) limit, even though the cc-pV5Z results are already very close to this limit. Designating the R-dependent potential energy function obtained with a given method and the cc-pVXZ basis set by $U_x(R)$, where $x=2,3,4$, and 5 correspond, respectively, to X = D, T, Q, and 5, we can estimate the cbs limit U_{cbs}, $U_{\mathrm{cbs}}(R)=\lim_{x\to\infty}U_x(R)$, by relying on a suitable extrapolation function. One possible choice due to Peterson and Dunning[43] is

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$$U_x(R) = U_{\rm cbs}(R) + B(R)e^{-(x-1)} + C(R)e^{-(x-1)^2}, \qquad (4)$$

where B and C are now geometry-dependent parameters.

Since cc-pV5Z potential is available only within a limited range of internuclear separations R, the resulting cbs extrapolated PEC covers the same range. The corresponding 4R-RMR/cbs vibrational levels are given in the second last column of Table VI, indicating that the cbs PEC yields a slightly better results than does the cc-pV5Z PEC.

It is important to emphasize that in contrast to standard extrapolations for a single property, here we extrapolate pointwise the entire PEC. Remarkably enough the B and C parameters in Eq. (4) change very little with R. Nonetheless, the extrapolation is not without problems, primarily due to a large change in the potential when going from a double-zeta to a triple-zeta basis. This is the reason why for the low-lying levels the extrapolated term values slightly deteriorate in the cbs limit (compare the 6th and 7th columns in Table VI). When we employ only the X = T, Q, and 5 PECs in the extrapolation, or when we use an additional term $D(R)\exp[-(x-1)^3]$ on the right-hand-side of the extrapolation function, Eq. (4), we obtain an improved agreement for the low-lying vibrational levels $(0<\nu<5)$. In fact, such extrapolations yield the term values that are very close to those obtained with the cc-pV5Z PEC.

The above mentioned behavior is not surprising in view of the difficulties that are encountered in the construction of the correlation-consistent (cc) basis sets for the studied system. In fact, the original paper of Dunning[44] gives no basis set for the Li atom. Nonetheless, for the higher-lying vibrational levels all the cbs extrapolations mentioned above represent an improvement. On the average, the extrapolated cbs data for the 19 lowest-lying levels are better than the cc-pV5Z ones by 8 to 9 cm^{-1}.

F. Rovibrational transitions

Recently, Dulick *et al.*[2] measured hundreds of rovibrational lines for the isotopomers ^{6}LiH(D) and ^{7}LiH(D) using the high-resolution Fourier transform spectra. For obvious reasons we do not present here the results for all the available lines and all isotopomers, but illustrate the performance of our results by selecting only four lines for each vibrational band, namely those characterized by the smallest and the largest rotational quantum number J in the observed P and R branches (in some cases, no R branch was observed).

Let us first consider ^{7}LiH for which the observed spectrum is the richest, i.e., in which case the largest number of vibrational bands have been recorded.[2] For the selected lines (see above) we list in Table IX the experimental transition frequencies and the deviations of the theoretical results obtained with the 4R-RMR method and the cc-pVXZ basis sets (X = D, T, Q, and 5). The CCSD-[4R] results are slightly better than the 4R-RMR ones (by only 0.1 cm^{-1} for a few lines listed in Table IX). We observe a consistent and systematic improvement in the computed frequencies as the size of the basis set increases from X = D, to X = T, to X = Q, and finally to X = 5. The average 4R-RMR errors decrease from more than 30 cm^{-1} for X = D, to 5.7 cm^{-1} for X = T, to 2 cm^{-1} for X = Q, to finally 1.5 cm^{-1} for X = 5 (cf. Table X). Thus, at the 4R-RMR/cc-pV5Z level, the agreement between the theory and experiment is very good indeed. The inclusion of the adiabatic corrections improves the 4R-RMR/cc-pV5Z results by less than 1 cm^{-1}, i.e., by about 0.5 cm^{-1} on the average.

TABLE X. Mean absolute deviations (in cm^{-1}) between the computed and experimental frequencies for 38 selected lines (cf. Table VII) in the infrared spectrum of ^{7}LiH, as obtained with various methods and cc-pVXZ (X=D, T, Q, and 5) basis sets.

Method	X:=D	T	Q	5
4R-RMR	−30.19	5.68	2.01	1.54
CCSD-[4R]	−30.20	5.66	1.98	
CCSD	−30.11	5.96	2.36	
4R-CISD	−29.65	12.48	15.95	
FCI	−30.31	5.34		

As an illustration of the performance of various methods that we employed in this study we present in Table X the average errors obtained for the 38 selected lines of ^{7}LiH that are listed in Table IX. We find that with the exception of 4R-CISD, the larger the basis set employed, the better are the results, and as we approach the cbs limit, the computed values approach the experimental data. With the cc-pVQZ basis, the 4R-RMR and CCSD-[4R] average errors are only 2 cm^{-1}. The differences between the cc-pVQZ and cc-pV5Z are already very small. Since the rovibrational transitions listed in Table IX involve at most the $\nu=6$ vibrational level, and the standard CCSD method only slightly underperforms the 4R-RMR or CCSD-[4R] approaches for the $\nu=0-6$ levels as already pointed out in Sec. III A, the CCSD/cc-pVQZ results for the 38 selected transitions differ from the CCSD-[4R] values by only a fraction of a wave number. We also note that for the smaller cc-pVDZ and cc-pVTZ basis sets, the 4R-RMR and CCSD-[4R] results are only slightly worse than the FCI values (again, by only a fraction of a wave number).

G. Other isotopomers

We can employ the obtained mass-independent, Born–Oppenheimer potential, with or without the mass-dependent adiabatic corrections, to calculate the vibrational levels for other isotopic LiH species. Since such results do not provide us with a substantially new information concerning the reliability of theoretically generated potentials, we will mention only briefly the results obtained at the CCSD-[4R]/cc-pVQZ level (without adiabatic corrections). Comparing with the experimentally available values from Ref. 1, we find for the 28 vibrational levels of ^{7}LiD the largest deviation to be 36.2 cm^{-1} and the average error 16.9 cm^{-1}. For the 22 levels of ^{6}LiH, the largest and the average deviations are 38.3 and 18.0 cm^{-1}, respectively. Finally, for the 10 levels of ^{6}LiD, the corresponding values are 7.1 are 3.2 cm^{-1}.

In order to make a comparison with the directly observed experimental data, we present in Table XI the experimental transition frequencies for the isotopomers ^{6}LiH(D) and ^{7}LiD and compare them with the computed values ob-

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TABLE XI. The discrepancies δ_X (X = 4 and 5) between the computed and experimental (Ref. 2) (Expt.) transition energies (as in Table IX) for the selected lines of three isotopomers of LiH, namely ^{6}LiH, ^{6}LiD, and ^{7}LiD, as obtained with the 4R-RMR CCSD/cc-pVXZ, X = 4 (or Q) and 5, potential including the adiabatic corrections. All values are in cm^{-1}.

	Band	Line	Expt.[a]	δ_4	δ_5	Line	Expt.[a]	δ_4	δ_5
^{6}LiH	(1,0)	P(20)	1016.4	3.3	0.4	P(1)	1358.3	1.5	0.7
		R(1)	1402.3	1.3	0.7	R(13)	1530.4	1.6	0.1
	(2,1)	P(20)	980.3	2.0	0.6	P(1)	1312.9	0.3	0.0
		R(0)	1341.8	0.2	−0.1	R(12)	1472.4	0.3	−0.2
	(3,2)	P(17)	1000.0	1.3	1.1	P(6)	1191.6	0.1	0.3
		R(3)	1335.3	−0.2	0.1	R(10)	1407.4	0.0	0.2
^{6}LiD	(1,0)	P(25)	791.7	2.7	0.4	P(3)	1020.9	1.3	0.7
		R(3)	1080.3	1.2	0.6	R(13)	1146.1	1.3	0.4
	(2,1)	P(25)	770.6	1.7	0.3	P(3)	995.1	0.5	0.2
	(3,2)	P(22)	781.9	1.0	0.5	P(6)	942.8	0.1	0.0
^{7}LiD	(1,0)	P(26)	772.0	2.7	0.4	P(1)	1020.7	1.3	0.7
		R(0)	1037.3	1.2	0.7	R(24)	1165.5	1.6	−0.1
	(2,1)	P(24)	772.8	1.6	0.3	P(1)	995.4	0.4	0.1
		R(0)	1011.6	0.4	0.1	R(21)	1129.0	0.6	−0.3
	(3,2)	P(21)	783.2	0.9	0.4	P(1)	970.5	0.0	−0.1
		R(0)	986.4	−0.1	−0.1	R(20)	1097.8	0.2	−0.1
	(4,3)	P(21)	762.6	0.8	0.7	P(1)	946.0	−0.2	0.1
		R(1)	969.0	−0.2	0.0	R(16)	1056.0	0.0	0.2
	(5,4)	P(18)	771.4	0.7	0.8	P(2)	914.0	0.0	0.4
		R(2)	951.5	−0.1	0.3	R(8)	990.1	0.0	0.3
	(6,5)	P(15)	778.8	0.8	0.9	P(9)	832.7	0.5	0.7

[a]Reference 2.

tained with the 4R-RMR method using the cc-pVQZ and cc-pV5Z basis sets. The adiabatic corrections are also included and for each P or R branch in a given band, we present the result for only two lines, namely those that are associated with the highest and the lowest J values.[45] Again, we find an excellent agreement between the theory and experiment. With the cc-pVQZ basis, the average deviations are in the range of 1 to 2 cm^{-1} for each isotopomer, while with the cc-pV5Z basis these deviations do not exceed 1 cm^{-1} (except in one case). Moreover, it should be noted that the discrepancies between the computed and experimental frequencies for the observed P and R lines that are associated with the smallest and the largest J quantum number are, correspondingly, the smallest and the largest ones (in absolute value). In other words, these deviations define the interval which encompasses the discrepancies for all other observed transitions that are associated with the intermediate J quantum numbers.

H. Potential energy curves

In concluding this paper, we present in Table XII the LiH PEC obtained with the CCSD-[4R]/cc-pVQZ method, which is very similar to the 4R-RMR CCSD/cc-pVQZ one, yet slightly better. One of the highly accurate PECs that have been extensively tabulated over a wide range of internuclear separations is that obtained by Partridge and Langhoff (PL),[3] who employed the CI method and Slater-type basis set. Nonetheless, this PEC seems to be still lacking the data for certains regions of the internuclear separations, since the energy differences for some neighboring points are too large. For example, the energy difference between the $R=5$ and $R=5.5$ bohr exceeds 2600 cm^{-1}. Likewise, the energy differences between $R=4.5$ and 4.75 bohr, as well as between $R=6.0$ and 6.5 bohr, are larger than 1500 cm^{-1}. Note that the outer turning point (R_+) for the $\nu=6$ level of the IPA curve falls right between $R=4.5$ and 4.75 bohr. Similarly, there are outer turning points for the three vibrational levels between 5.0 and 5.5 bohr. With so large energy gaps, the errors of several tens of cm^{-1} are possible when carrying out the vibrational analysis. Idealy, more points should be calculated in between the two geometries for which the energy gap is too large.

It is worth mentioning in this context that Lee *et al.*[7] recalculated the vibrational level spacings $\Delta G_\nu = G_{\nu+1} - G_\nu$ using the PEC of Partridge and Langhoff.[3] Their results[7] significantly differ from those given by Partridge and Langhoff.[3] For example, for the $7\rightarrow 8$ vibrational transition, Lee *et al.*[7] obtained the value 1070.3 cm^{-1}, while Partridge and Langhoff [3] found 1050.9 cm^{-1}, i.e., differing by almost 20 cm^{-1}. We have recalculated this spacing with the LEVEL codes of LeRoy[35] and found the value that is closer to that given by Partridge and Langhoff,[3] differing by less than 5 cm^{-1}. However, since the turning points for levels 7 and 8 are within the interval $R\in[4.5,5.5]$ bohr, where the tabulated PEC of Ref. 3 does not have enough points, larger errors are possible when carrying out the numerical analysis of the vibrational term values and corresponding transition energies.

IV. CONCLUSIONS

The availability of highly accurate spectroscopic data for LiH and its various isotopomers,[1,2] as well as of the vibrational energy levels and the potential energy curve (PEC) deduced via a careful analysis of these data[36] using the in-

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TABLE XII. The CCSD-[4R]/cc-pVQZ energies (in a.u.) of LiH for various internuclear separations R (in Å), defining the potential energy curve that was used in computations of the corresponding rovibronic energy levels.

R	CCSD-[4R]	R	CCSD-[4R]
1.0	−7.962 968 6	3.7	−7.974 508 5
1.1	−8.001 001 5	3.8	−7.972 844 8
1.2	−8.026 067 4	3.9	−7.971 441 2
1.3	−8.041 909 7	4.0	−7.970 269 0
1.4	−8.051 198 5	4.1	−7.969 298 5
1.5	−8.055 813 9	4.2	−7.968 500 7
1.6	−8.057 089 5	4.3	−7.967 848 7
1.7	−8.055 996 9	4.4	−7.967 318 5
1.8	−8.053 255 2	4.5	−7.966 889 0
1.9	−8.049 396 2	4.6	−7.966 542 2
2.0	−8.044 807 2	4.7	−7.966 262 9
2.1	−8.039 766 2	4.8	−7.966 038 4
2.2	−8.034 471 8	4.9	−7.965 858 4
2.3	−8.029 069 8	5.0	−7.965 714 4
2.4	−8.023 670 7	5.1	−7.965 599 4
2.5	−8.018 359 5	5.2	−7.965 507 7
2.6	−8.013 200 4	5.3	−7.965 434 9
2.7	−8.008 241 1	5.4	−7.965 377 2
2.8	−8.003 517 0	5.5	−7.965 331 8
2.9	−7.999 055 2	5.6	−7.965 296 2
3.0	−7.994 877 1	5.7	−7.965 268 5
3.1	−7.991 000 0	5.8	−7.965 247 1
3.2	−7.987 437 8	5.9	−7.965 231 1
3.3	−7.984 200 5	6.0	−7.965 218 9
3.4	−7.981 293 1	6.2	−7.965 203 7
3.5	−7.978 714 9	6.4	−7.965 196 8
3.6	−7.976 458 3	6.6	−7.965 194 9

verted perturbation approach[37–39] (IPA), enables a critical assessment of the theoretically determined potentials and thus of the performance and reliability of various post-HF quantum chemical methods. In this paper we have focused on the performance of the externally corrected (ec) CCSD approaches of both the amplitude- and energy-correcting kind, namely the RMR CCSD and CCSD-[MR] methods, vis-à-vis the MR CISD—that was used as the external source—and the standard SR CCSD methods. A comparison with the existing PECs obtained by other methods, including the state-universal MR CCSD (Ref. 8) and CCSD/BO (Ref. 7) methods, has also been made as long as the corresponding data were available.

In order to assess the role of the basis set error vs the error of the method employed, we have exploited the fact that in the case of the LiH molecule we deal with only a four-electron system, in which case it was feasible to generate the FCI energies using an intermediately large cc-pVTZ basis set that enables one to generate realistic results that may be compared with experiment. This comparison clearly showed that the main errors at this level are due to the basis set inadequacy, since the approximate 4R-RMR CCSD and CCSD-[4R] energies differ very little from the FCI ones. This is a very encouraging result which confirms our earlier comparisons of the approximate and FCI energies that were, however, carried out only for the cc-pVDZ models in view of a larger number of electrons involved.

Of course, in the LiH case the SR CCSD performs already well in view of the fact that the quasidegeneracy effects associated with the stretching of the Li–H bond become prominent only in the immediate vicinity of the dissociation limit, thus significantly influencing only a few highest-lying vibrational levels. We have thus focused our attention on these highest-lying levels that enable us to make a critical assessment of the various methods mentioned above.

Our results clearly demonstrate an excellent performance of both ecCCSD methods that employ a modest-size MR CISD wave function as the source of higher-than-pair clusters, thus enabling a very accurate and reliable *ab initio* determination of the entire manifold of the rovibrational frequencies with a very high accuracy (cf. Table IX). While the errors for the individual vibrational levels may lie in the range of 10–30 cm^{-1}, the discrepancies between the experimentally observable and computed transition energies never exceed couple of wave numbers, and for the most transitions do not exceed 1 cm^{-1}. Moreover, these deviations systematically increase with the increasing vibrational and rotational quantum numbers, so that the computed rovibrational spectrum matches extremely well the experimental one.

We have also examined the dissimilarity between the results obtained with Cartesian and spherical basis functions. A generally accepted belief purports that the use of spherical Gaussians reduces the computational cost without sacrificing the performance. Our results clearly indicate that when exploring the entire potential at a highly correlated level, the Cartesian and spherical basis functions lead to significantly distinct PECs and vibrational energies. In fact, these differences are larger than the multireference effects, which are the main focus of this study. The results obtained with Cartesian Gaussians, at least for the system studied here, are definitely superior to those obtained with corresponding spherical Gaussians. This is undoubtedly due to the fact that the Cartesian bases are considerably larger—and thus more flexible—than the spherical ones, since they implicitly contain AOs with smaller angular momentum quantum numbers.

We can thus conclude that both ecCCSD approaches, which yield remarkably similar results, are clearly superior to the corresponding modest-size MR CISD method that is employed as the external source of nondynamical correlation effects, and represent a significant improvement over the standard SR CCSD and even over the state-universal MR CCSD approaches.[8] Thus, a small reference space RMR CCSD or CCSD-[4R] approaches can provide as good results as a large reference space MR CISD.

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